

Polyhedral angles and polyhedra

What happens at a vertex of a solid — the angle condition every corner must satisfy.

LESSON 19–20

2 × 40 MIN

GEOMETRY 11, §9

IGCSE E13.X

LESSON CLOCK

12'

22'

22'

20'

18'

6'

What a polyhedral angle is	12 min
The angle inequality	22 min
Convexity	22 min
Euler again	20 min
Practice	18 min
Homework	6 min

LEARNING OBJECTIVES

- Define a polyhedral angle and name its plane and dihedral angles.
- Apply the inequality on the plane angles at a vertex.
- Classify polyhedra as convex or non-convex.
- Use Euler’s formula on a given solid.

TEXTBOOK

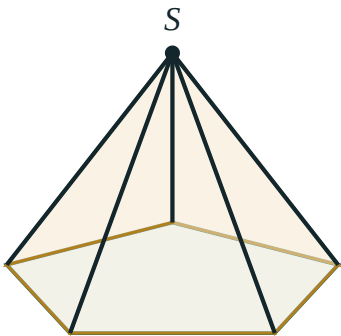
National	pp. 89–100
Cambridge	Core & Extended, pp. 282–288
Workbook	Exercise 9.1

01

Explanation

The corner of a solid

Take a point S and several rays from it, no three coplanar, joined in order by plane faces. The figure is a **polyhedral angle**: three rays give a **trihedral** angle, four a tetrahedral one, and so on.



five plane angles at S , five dihedral angles along the edges
the sum of the plane angles is less than 360°

Five faces meeting at S . Each face contributes a plane angle; each edge a dihedral angle.

TWO KINDS OF ANGLE AT A CORNER

A **plane angle** lies inside one face, between two adjacent edges. A **dihedral angle** is between two adjacent faces, measured along their common edge. A corner with n edges has n of each.

The angle inequality

TWO THEOREMS ABOUT THE PLANE ANGLES

1 Each plane angle is less than the sum of the others. **2** The sum of all the plane angles at a vertex is less than 360° .

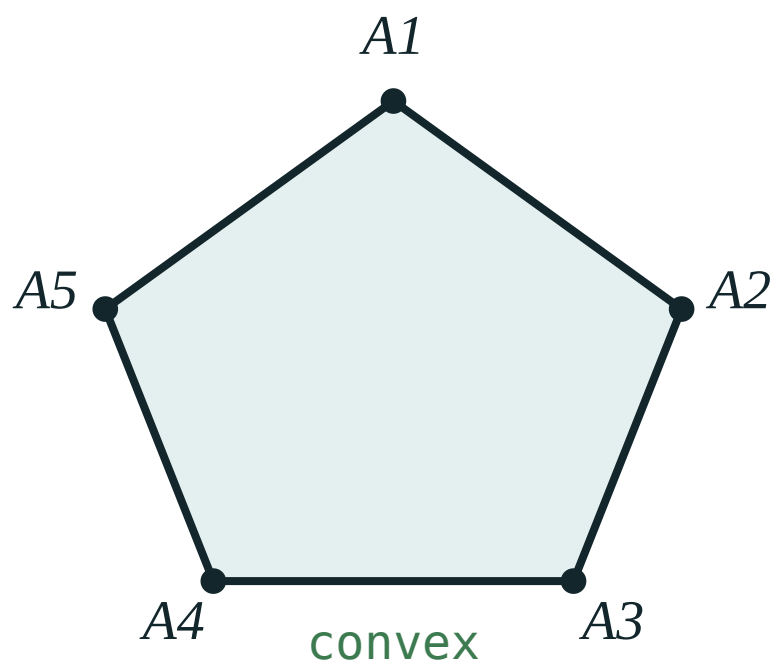
Theorem 2 is the reason there are only five regular polyhedra: with equilateral triangles (60°) at most five fit; with squares (90°) at most three; with regular pentagons (108°) at most three; six hexagons give exactly 360° and lie flat, forming no corner at all.

EXACTLY 360° IS NOT A CORNER

It is a flat piece of plane. To fold up into a solid, a corner must have angles totalling **strictly less** than a full turn — the deficit is what makes it pop out.

Convexity

A polyhedron is **convex** when it lies entirely on one side of the plane of each of its faces — equivalently, when the segment joining any two of its points stays inside.



Convex on the left. On the right, a segment between two interior points leaves the solid.

PROPERTY	CONVEX	NON - CONVEX
Euler's formula $F + V - E = 2$	always	usually, but not for a solid with a hole
every section is a single polygon	yes	no
every plane angle sum $< 360^\circ$	yes	can exceed it at a reflex corner

Euler again, as a tool

$F + V - E = 2$

Two counting facts make it usable when the faces are known:

$2E = \Sigma (\text{edges per face}) = \Sigma (\text{edges per vertex})$

A solid with 12 pentagons and 20 hexagons, three faces at each vertex: $2E = 12 \times 5 + 20 \times 6 = 180$, so $E = 90$; and $3V = 2E = 180$ gives $V = 60$. Check: $32 + 60 - 90 = 2$
✓. That solid is the football.

02 Worked examples

EXAMPLE 1 Can a corner have three plane angles of 120° each?

- ① $3 \times 120^\circ = 360^\circ$
- ② The sum must be strictly less.
- ③ They lie flat.

ANSWER

No — it is a flat plane, not a corner

EXAMPLE 2 A corner has plane angles 70° , 80° and x . Find the possible values of x .

- ① Sum $< 360^\circ$: $x < 210^\circ$.
- ② Each less than the sum of the others: $x < 150^\circ$.
- ③ And $80 < 70 + x$, so $x > 10^\circ$.

ANSWER

$10^\circ < x < 150^\circ$

EXAMPLE 3 A polyhedron has 12 pentagonal and 20 hexagonal faces, 3 at each vertex. Find V and E .

① $2E = 12 \times 5 + 20 \times 6 = 180$
 $E = 90$

② $3V = 2E = 180$
 $V = 60$

③ Check: $32 + 60 - 90 = 2$ ✓

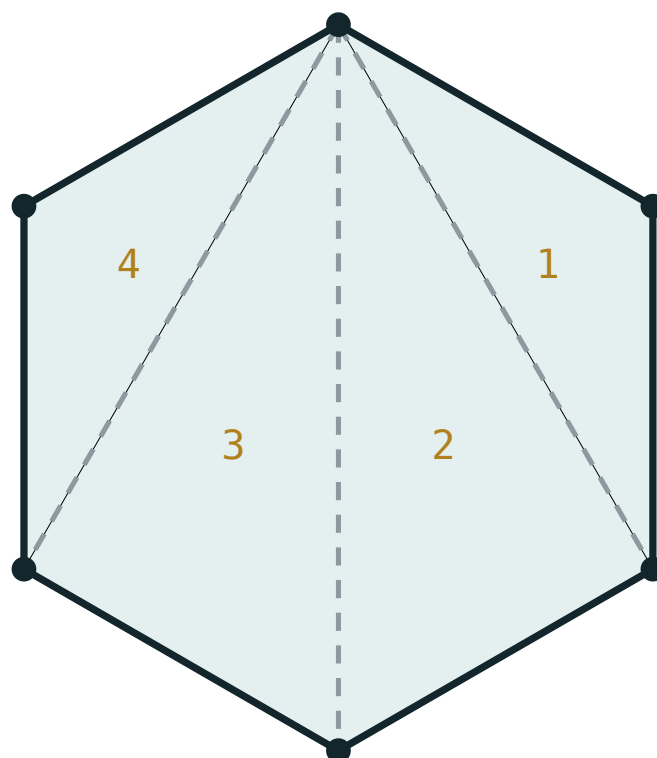
ANSWER

$$V = 60, E = 90$$

03 Interactive model

Fold three squares at a corner and try to add a fourth — the angles refuse.

Angle sum of a polygon



$$180^\circ \times (6 - 2) = 720^\circ$$

sides n **6**

triangles **4**

interior sum **720°**

each angle **120°**

exterior sum **360°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Polyhedral angle	Ko'pyoqli burchak	Многогранный угол
Trihedral angle	Uch yoqli burchak	Трёхгранный угол
Plane angle (of a corner)	Yassi burchak	Плоский угол
Dihedral angle	Ikki yoqli burchak	Двугранный угол
Apex (vertex)	Uchi	Вершина
Convex polyhedron	Qavariq ko'pyoq	Выпуклый многогранник
Non-convex	Qavariq bo'lmagan	Невыпуклый
Euler's formula	Eyler formulasi	Формула Эйлера
Lateral face	Yon yoq	Боковая грань

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The plane angles at a vertex sum to:

A trihedral angle has how many faces?

Three regular hexagons at a corner:

Euler's formula holds for:

every solid

every convex polyhedron

only prisms

only regular solids

$2E$ equals:

the number of faces

the sum of edges per face

$F + V$

$3V$ always

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. How many faces has a trihedral angle?

ANSWER 3

2. Sum of plane angles at a vertex is

ANSWER less than 360°

3. Can three 120° angles form a corner?

ANSWER no

4. Can three 90° angles?

ANSWER yes — a cube corner

5. Can five 60° angles?

ANSWER yes — an icosahedron corner

6. Can six 60° angles?

ANSWER no — exactly 360°

7. State Euler's formula

ANSWER $F + V - E = 2$

Medium · the standard question

7 PROBLEMS

1. Plane angles 70° , 80° , x . Find the range of x

ANSWER $10^\circ < x < 150^\circ$

2. $F = 8$, $V = 6$; find E

ANSWER 12

3. $F = 20$, $E = 30$; find V

ANSWER 12

4. 12 pentagons + 20 hexagons, 3 per vertex: V , E

ANSWER 60, 90

5. A solid with only triangular faces and $V = 6$, 4 per vertex: E , F

ANSWER 12, 8

6. Why are there only five regular polyhedra?

ANSWER The angle sum at a corner must be under 360°

7. Is a picture-frame solid convex?

ANSWER no

Hard · stretch and reason

7 PROBLEMS

1. Prove each plane angle is less than the sum of the others

ANSWER Unfold the corner into the plane and use the triangle inequality

2. Prove the plane angles sum to less than 360°

ANSWER Project the corner onto a plane and compare

3. A convex solid has 7 faces. What is the least possible E ?

ANSWER 12 — a hexagonal pyramid

4. Show $2E \geq 3F$ and $2E \geq 3V$ for any polyhedron

ANSWER At least 3 edges per face and per vertex

5. Deduce $F \leq 2V - 4$ and $V \leq 2F - 4$

ANSWER Substitute both into Euler

6. Can a convex polyhedron have exactly 7 edges?

ANSWER no

7. A solid has faces that are all squares and triangles, 12 vertices, 4 edges at each. Find E , F

ANSWER $E = 24$, $F = 14$

07 Homework

Homework — 5 tasks

Every Euler answer must be checked against $F + V - E = 2$.

1. A corner has plane angles 60° , 100° and x . Find the range of x .
2. Explain, with the angle sums, why there are exactly five regular polyhedra.
3. A solid has 12 faces and 30 edges. Find V and name it if it is regular.
4. A solid has only triangular faces and 20 of them. Find E and V .

5. Give an example of a solid for which Euler's formula fails, and say why.

The prism, the parallelepiped and the cube

The family, its sections, and the diagonal formula that every cuboid problem reduces to.

LESSON 21–22

2 × 40 MIN

GEOMETRY 11, §10

IGCSE E13.X

LESSON CLOCK

12'

22'

24'

22'

16'

4

The family	12 min
Sections	22 min
The parallelepiped	24 min
The diagonal formula	22 min
Practice	16 min
Homework	4 min

LEARNING OBJECTIVES

- Name the elements of a prism and classify it as right, oblique or regular.
- Describe the sections of a prism parallel to the base and through a lateral edge.
- Prove the properties of a parallelepiped.
- Use the space-diagonal formula for a cuboid.

TEXTBOOK

National	pp. 101–112
Cambridge	Core & Extended, pp. 289–296
Workbook	Exercise 10.1

01

Explanation

The family

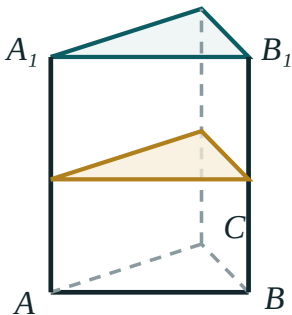
DEFINITION

A **prism** has two congruent parallel bases joined by parallelograms. It is **right** when the lateral edges are perpendicular to the bases, and **regular** when it is right and its base is a regular polygon.

SOLID	BASE	LATERAL FACES
oblique prism	any polygon	parallelograms
right prism	any polygon	rectangles
regular prism	regular polygon	congruent rectangles
parallelepiped	parallelogram	parallelograms
cuboid	rectangle	rectangles
cube	square	squares

Each row is a special case of the one above it. Every theorem proved for a prism therefore holds for a cube.

Sections



a section parallel
to the base is
congruent to it

every cross-section of a prism repeats the base
*A plane parallel to the base cuts a triangle congruent to
it — at every height.*

CUTTING PLANE	SECTION
parallel to the base	congruent to the base
through two lateral edges	a parallelogram (a diagonal section)
perpendicular to the lateral edges	the perpendicular section

The perpendicular section matters because the lateral surface area of **any** prism, right or oblique, is the perimeter of the perpendicular section times the lateral edge.

The parallelepiped

Four properties, each proved from the parallel-plane theorems of Grade 10:

- 1. Opposite faces are congruent parallelograms lying in parallel planes.
- 2. All four space diagonals meet at one point and bisect each other.
- 3. That point is the centre of symmetry of the solid.
- 4. In a **cuboid** all four space diagonals are equal.

Property 2 says the parallelepiped is the space analogue of the parallelogram, whose diagonals also bisect each other. The proof is the same: the two diagonals of any diagonal section are the diagonals of a parallelogram.

The diagonal formula

$$d^2 = a^2 + b^2 + c^2$$

Pythagoras twice — once in the base to get $\sqrt{a^2 + b^2}$, once vertically. For a cube of edge a it gives $d = a\sqrt{3}$.

CUBOID	BASE DIAGONAL	SPACE DIAGONAL
$3 \times 4 \times 12$	5	13
$1 \times 2 \times 2$	$\sqrt{5}$	3
$2 \times 3 \times 6$	$\sqrt{13}$	7
$a \times a \times a$	$a\sqrt{2}$	$a\sqrt{3}$

THREE SQUARES, NOT TWO

Every edge contributes. Learners who use only the two base edges get the **face** diagonal and answer a different question.

02 Worked examples

EXAMPLE 1 A cuboid measures $6 \times 8 \times 10$. Find its space diagonal and its four diagonal sections.

$$① \quad d = \sqrt{36 + 64 + 100} = \sqrt{200}$$

$$② \quad = 10\sqrt{2} \approx 14.1$$

$$③ \quad \text{Diagonal sections are rectangles } 10 \times \sqrt{100} \text{ and } 10 \times \sqrt{136}.$$

Two different sizes.

ANSWER

$$10\sqrt{2} \approx 14.1$$

EXAMPLE 2 A regular hexagonal prism has base edge 4 and height 9. Find the longest diagonal.

$$① \quad \text{The longest base diagonal of a regular hexagon is } 2 \times 4 = 8.$$

$$② \quad d = \sqrt{64 + 81}$$

$$③ \quad = \sqrt{145} \approx 12.04$$

ANSWER

$$\sqrt{145} \approx 12.0$$

EXAMPLE 3 A cube has space diagonal $9\sqrt{3}$. Find its edge and its face diagonal.

① $a\sqrt{3} = 9\sqrt{3}$
 $a = 9$

② Face diagonal $a\sqrt{2} = 9\sqrt{2}$

③ ≈ 12.7

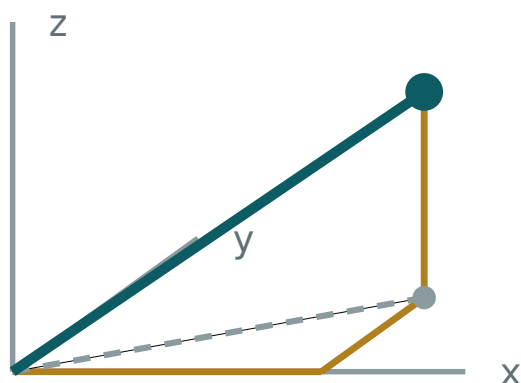
ANSWER

Edge 9, face diagonal $9\sqrt{2}$

03 Interactive model

Hold a box and trace the base diagonal, then the space diagonal, saying the two Pythagoras steps.

The diagonal of a box



x 3

y 2

z 2

OM 4.12

base diagonal 3.61

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Prism	Prizma	Призма
Right prism	To'g'ri prizma	Прямая призма
Oblique prism	Og'ma prizma	Наклонная призма
Regular prism	Muntazam prizma	Правильная призма
Parallelepiped	Parallelepiped	Параллелепипед
Cuboid	To'g'ri burchakli parallelepiped	Прямоугольный параллелепипед
Lateral edge	Yon qirra	Боковое ребро
Diagonal section	Diagonal kesim	Диагональное сечение
Space diagonal	Fazoviy diagonal	Пространственная диагональ
Perpendicular section	Perpendikulyar kesim	Перпендикулярное сечение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A right prism has lateral faces that are:

parallelograms

rectangles

triangles

trapezia

A section parallel to the base is:

smaller

congruent to the base

a triangle

a point

The space diagonal of an $a \times b \times c$ cuboid is:

$a + b + c$

$\sqrt{a^2 + b^2}$

$\sqrt{a^2 + b^2 + c^2}$

abc

The diagonals of a parallelepiped:

are equal

bisect each other

are perpendicular

do not meet

A cube of edge 5 has space diagonal:

$5\sqrt{2}$

$5\sqrt{3}$

15

25

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Space diagonal of a $3 \times 4 \times 12$ cuboid

ANSWER 13

2. Space diagonal of a cube of edge 5

ANSWER $5\sqrt{3}$

3. Face diagonal of a cube of edge 5

ANSWER $5\sqrt{2}$

4. Lateral faces of a right prism are

ANSWER rectangles

5. Section parallel to the base is

ANSWER congruent to the base

6. How many space diagonals has a cuboid?

ANSWER 4

7. Space diagonal of a $1 \times 2 \times 2$ cuboid

ANSWER 3

Medium · the standard question

7 PROBLEMS

1. Space diagonal of a $6 \times 8 \times 10$ cuboid

ANSWER $10\sqrt{2}$

2. Cube with space diagonal $9\sqrt{3}$; find the edge

ANSWER 9

3. Regular hexagonal prism, base edge 4, height 9: longest diagonal

ANSWER $\sqrt{145}$

4. Cuboid $2 \times 3 \times 6$; space diagonal

ANSWER 7

5. A cuboid has square base 5 and diagonal 13. Find the height

ANSWER $\sqrt{119} \approx 10.9$

6. Diagonal section of a cube of edge 6: its area

ANSWER $36\sqrt{2}$

7. Perpendicular section of a right prism is

ANSWER congruent to the base

Hard · stretch and reason

7 PROBLEMS

1. A cuboid has edges in ratio $1:2:3$ and diagonal $\sqrt{56}$. Find them

ANSWER 2, 4, 6

2. Prove the four space diagonals of a parallelepiped meet at one point

ANSWER Each diagonal section is a parallelogram

3. Oblique prism, lateral edge 10, perpendicular-section perimeter 18. Lateral area

ANSWER 180

4. A cube of edge a : angle between the space diagonal and a face diagonal

ANSWER $\approx 35.26^\circ$

5. A cuboid with all four space diagonals equal must be

ANSWER any cuboid — they always are

6. A parallelepiped whose diagonals are all equal is

ANSWER a cuboid

7. A regular triangular prism, base edge 6, height 8: longest diagonal

ANSWER 10

07 Homework

Homework — 5 tasks

Every diagonal answer must show both Pythagoras steps.

- Find the space diagonal of a $5 \times 12 \times 84$ cuboid.
- A cube has space diagonal 12. Find its edge and its face diagonal.
- A regular hexagonal prism has base edge 5 and height 12. Find its longest diagonal.
- Prove that the four space diagonals of a cuboid are equal.
- A cuboid has edges in the ratio $2 : 3 : 6$ and space diagonal 14. Find the edges.

The surface area of a prism

Two bases and one rectangle — the net says the formula better than the formula does.

LESSON 23

1 × 40 MIN

GEOMETRY 11, §11

IGCSE E13.X

LESSON CLOCK

8'

12'

10'

10'

5'

The net	8 min
The formula	12 min
Oblique prisms	10 min
Working backwards	10 min
Homework	5 min

LEARNING OBJECTIVES

- Derive the lateral surface area of a right prism from its net.
- Compute the total surface area of a prism.
- Use the perpendicular section for an oblique prism.
- Solve for a missing dimension from a given surface area.

TEXTBOOK

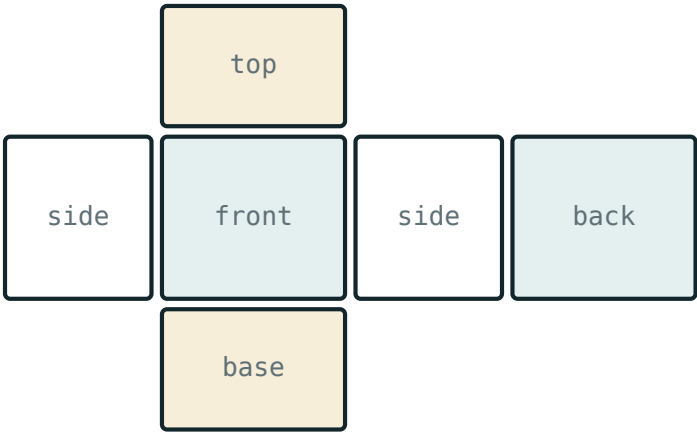
National	pp. 113–118
Cambridge	Core & Extended, pp. 297–300
Workbook	Exercise 11.1

01

Explanation

The net

Cut a right prism along one lateral edge and roll the sides flat. The lateral surface is a single rectangle: its height is the prism's height, and its width is the base perimeter.



surface area = the area of the whole net

The lateral surface unrolls into one rectangle, P wide and h tall.

$$S_{lat} = P \cdot h \quad S_{total} = 2B + P h$$

where B is the base area and P the base perimeter.

Worked shapes

PRISM	B	P	S_{TOTAL}
cuboid $a \times b \times h$	ab	$2(a + b)$	$2ab + 2(a+b)h$
cube, edge a	a^2	$4a$	$6a^2$
regular triangular, edge a , height h	$\frac{\sqrt{3}}{4} a^2$	$3a$	$\frac{\sqrt{3}}{2} a^2 + 3ah$
regular hexagonal, edge a , height h	$\frac{3\sqrt{3}}{2} a^2$	$6a$	$3\sqrt{3} a^2 + 6ah$

TWO BASES, NOT ONE

A closed prism has a lid. $S_{total} = 2B + Ph$: the 2 is the mark most often lost. An open container — a trough, a box without a lid — needs the formula adjusted, and the question will say so.

Oblique prisms

For an oblique prism the lateral faces are parallelograms of different shapes, and the net is not one rectangle. The formula generalises by using the **perpendicular section** — the cut at right angles to the lateral edges:

$$S_{lat} = P_{\perp} \cdot \ell \text{ where } \ell \text{ is the lateral edge}$$

For a right prism the perpendicular section is the base and $\ell = h$, so the general formula reduces to the one above.

02 Worked examples

EXAMPLE 1 A regular hexagonal prism has base edge 4 cm and height 10 cm. Find its total surface area.

$$① \quad B = \frac{3\sqrt{3}}{2} \times 16 = 24\sqrt{3} \approx 41.57$$

$$② \quad P = 24$$

$$③ \quad S_{lat} = 240$$

$$④ \quad S = 2(41.57) + 240 \approx 323.1$$

ANSWER

$$48\sqrt{3} + 240 \approx 323 \text{ cm}^2$$

EXAMPLE 2 A cuboid has surface area 136 cm^2 and base 4×5 . Find its height.

$$① \quad 2(20) + 2(9)h = 136$$

$$② \quad 18h = 96$$

$$③ \quad h = \frac{16}{3} \approx 5.33$$

ANSWER

$$\frac{16}{3} \approx 5.33 \text{ cm}$$

EXAMPLE An oblique prism has lateral edge 12 and perpendicular section a triangle with sides 3, 4, 5. Find the lateral area.

① $P_{\perp} = 12$

② $S_{lat} = 12 \times 12$

③ $= 144$

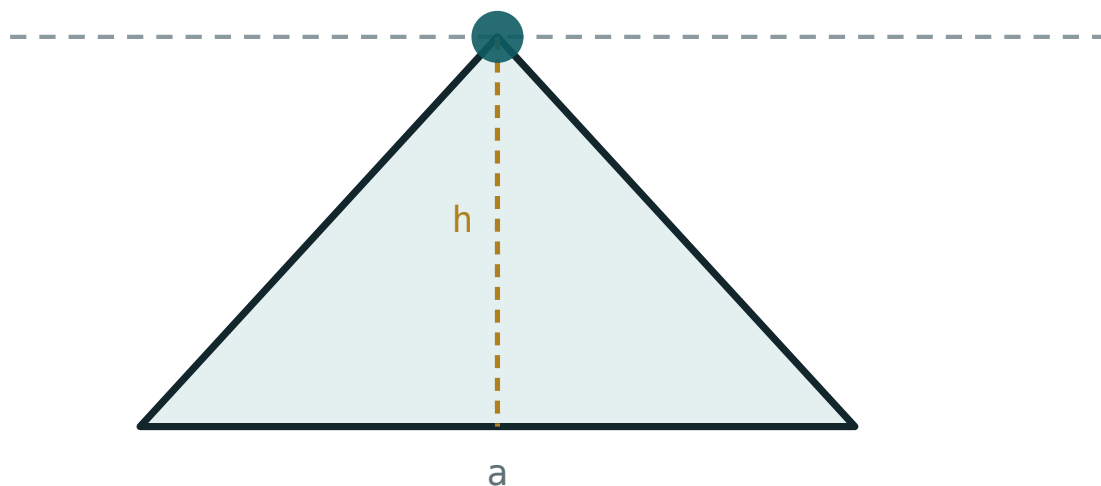
ANSWER

144 square units

03 Interactive model

Unroll a cardboard tube or box and measure the rectangle directly.

Area from a net

base a **220**height h **120**area $\frac{1}{2}ah$ **13200**perimeter **545.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Lateral surface area	Yon sirt yuzasi	Площадь боковой поверхности
Total surface area	To'la sirt yuzasi	Площадь полной поверхности
Net	Yoyma	Развёртка
Base perimeter	Asos perimetri	Периметр основания
Base area	Asos yuzasi	Площадь основания
Perpendicular section	Perpendikulyar kesim	Перпендикулярное сечение
Lateral edge	Yon qirra	Боковое ребро
Apothem of a polygon	Ko'pburchak apofemasi	Апофема многоугольника

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

S_{lat} of a right prism is:

S_{total} of a right prism is:

A cube of edge 4 has surface area:

For an oblique prism S_{lat} uses:

the base perimeter

the perpendicular section perimeter

the height

the diagonal

An open box needs:

$2B + Ph$

$B + Ph$

Ph

$2B$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Surface area of a cube of edge 3

ANSWER 54

2. Surface area of a cube of edge 7

ANSWER 294

3. Cuboid $2 \times 3 \times 4$

ANSWER 52

4. Cuboid $5 \times 5 \times 10$

ANSWER 250

5. S_{lat} of a prism with $P = 20$, $h = 6$

ANSWER 120

6. S_{total} with $B = 15$, $P = 16$, $h = 5$

ANSWER 110

7. A cube has surface area 150. Find its edge

ANSWER 5

Medium · the standard question

7 PROBLEMS

1. Regular triangular prism, edge 6, height 10: total area

ANSWER $18\sqrt{3} + 180 \approx 211$

2. Regular hexagonal prism, edge 4, height 10: total area

ANSWER $48\sqrt{3} + 240 \approx 323$

3. Cuboid with base 4×5 and area 136: the height

ANSWER $\frac{16}{3}$

4. Right prism, base a 3-4-5 triangle, height 9: total area

ANSWER $12 + 108 = 120$

5. Oblique prism, edge 12, perpendicular perimeter 12: lateral area

ANSWER 144

6. Open cuboid $3 \times 4 \times 5$ (no lid): area

ANSWER 82

7. A cube of surface area 384: its space diagonal

ANSWER $8\sqrt{3}$

Hard · stretch and reason

7 PROBLEMS

1. A cuboid has square base and total area 294 with height equal to the base edge. Find the edge

ANSWER 7

2. Regular hexagonal prism of base edge a and height a : total area

ANSWER $3\sqrt{3}a^2 + 6a^2$

3. A prism has a rhombus base with diagonals 6 and 8, height 10. Total area

ANSWER $48 + 200 = 248$

4. A cuboid of volume 240 and square base 4: total area

ANSWER 272

5. Two similar prisms with areas 75 and 300: the length ratio

ANSWER 1 : 2

6. A prism's lateral area equals its total area minus what?

ANSWER twice the base area

7. Prove $S_{lat} = P_{\perp} \ell$ for an oblique prism

ANSWER Each parallelogram has base a side of the perpendicular section

07 Homework

Homework — 4 tasks

Say whether the solid is closed or open before choosing the formula.

1. A regular hexagonal prism has base edge 6 cm and height 15 cm. Find its total surface area.
2. A cuboid with base 6×8 has total surface area 376 cm^2 . Find its height.
3. A right prism has a 5-12-13 triangular base and height 20. Find its total surface area.
4. An open trough is a cuboid $2 \times 3 \times 8$ with no lid. Find the area of material needed.

Upper and lower bounds in measurement

Cambridge insert: every measurement is an interval, and every calculation from it is an interval too.

LESSON 24

1 × 40 MIN

GEOMETRY 11, §11 (EXTENSION)

IGCSE E13.3

LESSON CLOCK

8'

12'

14'

8'

3'

A measurement is an interval	8 min
The four operations	12 min
Areas and volumes	14 min
How many figures	8 min
Homework	3 min

LEARNING OBJECTIVES

- Write the upper and lower bound of a rounded measurement.
- Find the bounds of a sum, difference, product and quotient.
- Apply bounds to the area and volume of a solid.
- Decide how many figures an answer can honestly carry.

TEXTBOOK

National	pp. 119–122
Cambridge	Core & Extended, pp. 44–48
Workbook	IGCSE Exercise 13.3

01

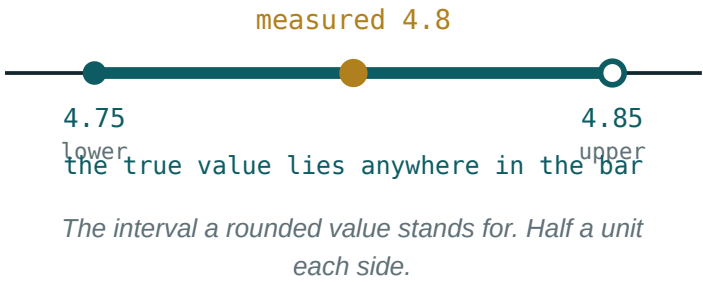
Explanation

A measurement is an interval

HALF THE ROUNDING UNIT, EACH WAY

A length given as 8 cm to the nearest cm lies in $7.5 \leq \ell < 8.5$. To the nearest 0.1 cm, 8.0 means $7.95 \leq \ell < 8.05$.

The lower bound is included and the upper is not — but at this level both are written as $\leq$, and the distinction is only mentioned.



The four operations

QUANTITY	GREATEST WHEN	LEAST WHEN
$a + b$	both are greatest	both are least
$a - b$	a greatest, b least	a least, b greatest
$a \times b$	both greatest	both least
$\frac{a}{b}$	a greatest, b least	a least, b greatest

SUBTRACTION AND DIVISION SWAP THE BOUNDS

To make a difference as large as possible, take the largest top and the **smallest** bottom. Using the upper bound of both is the standard error.

Areas and volumes

A cuboid measured as $6 \times 8 \times 10$ cm, each to the nearest cm, has

$$V_{max} = 6.5 \times 8.5 \times 10.5 = 580.125 \quad V_{min} = 5.5 \times 7.5 \times 9.5 = 391.875$$

The stated volume is 480 cm^3 , but the true value lies anywhere in a range of nearly 190 cm^3 . Small uncertainties in three lengths compound sharply in the product.

WHY THIS INSERT SITS HERE

Every surface-area and volume formula of this quarter multiplies two or three measured lengths. Bounds are how a real answer is stated, and the reason a measured result is never quoted to eight figures.

How many figures

Quote the answer only to the digits that agree between the upper and lower bounds. For the cuboid: $V_{min} = 391.9$, $V_{max} = 580.1$ — not even the first digit agrees, so the honest statement is “between about 390 and 580 cm^3 ”.

Measure the same box to the nearest millimetre and the bounds become 478.5 and 481.5 : now “ 480 cm^3 ” is justified.

02 Worked examples**EXAMPLE 1** A length is 12.4 cm to 1 d.p. Give its bounds.

① Rounding unit 0.1; half is 0.05.

② $12.35 \leq \ell \leq 12.45$

ANSWER

12.35 and 12.45 cm

EXAMPLE 2 A rectangle is 7 cm by 4 cm, each to the nearest cm. Find the bounds of its area.

① $6.5 \leq a \leq 7.5, 3.5 \leq b \leq 4.5$.

② $A_{\min} = 6.5 \times 3.5 = 22.75$

③ $A_{\max} = 7.5 \times 4.5 = 33.75$

ANSWER22.75 to 33.75 cm²

EXAMPLE
3

A journey of 150 km (to the nearest 10 km) takes 2.5 h (to the nearest 0.1 h). Find the greatest possible speed.

① Distance greatest: 155.

② Time least: 2.45.

③ $\frac{155}{2.45} \approx 63.3$

ANSWER $\approx 63.3 \text{ km/h}$

03 Interactive model

Measure the same object with a ruler and with a tape and compare the intervals.

Bounds

8 cm to the nearest cm has lower bound:

4.60 to 2 d.p. has upper bound:

To maximise $a - b$ take:

To maximise a b take:

both maxima

a max, b min

both minima

a min, b max

A 7×4 rectangle (nearest cm) has greatest area:

28

30

33.75

35

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Upper bound	Yuqori chegara	Верхняя граница
Lower bound	Quyí chegara	Нижняя граница
Rounded to the nearest	Yaxlitlangan	Округлено до
Degree of accuracy	Aniqlik darajasi	Степень точности
Interval of values	Qiymatlar oralig'i	Промежуток значений
Maximum value	Eng katta qiymat	Максимальное значение
Minimum value	Eng kichik qiymat	Минимальное значение
Significant figures	Muhim raqamlar	Значащие цифры

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

20 m to the nearest 10 m lies in:

19–21

15–25

19.5–20.5

10–30

A measurement is best described as:

an exact number

an interval

a guess

a fraction

Bounds of a volume from three measured lengths:

add

multiply the corresponding bounds

stay the same

halve

An answer should be quoted to:

as many figures as the calculator gives

the figures the bounds agree on

two figures always

one figure

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Bounds of 9 cm to the nearest cm

ANSWER 8.5 and 9.5

2. Bounds of 4.2 to 1 d.p.

ANSWER 4.15 and 4.25

3. Bounds of 300 to the nearest 100

ANSWER 250 and 350

4. Bounds of 12.40 to 2 d.p.

ANSWER 12.395 and 12.405

5. Greatest $a + b$ for $a = 5$, $b = 3$ (nearest 1)

ANSWER 9

6. Least $a + b$ for the same

ANSWER 7

7. Greatest $a - b$ for the same

ANSWER 3

Medium · the standard question

7 PROBLEMS

1. Rectangle 7×4 (nearest cm): area bounds

ANSWER 22.75 to 33.75

2. Cuboid $6 \times 8 \times 10$ (nearest cm): volume bounds

ANSWER 391.875 to 580.125

3. Speed: 150 km (nearest 10), 2.5 h (nearest 0.1). Greatest speed

ANSWER ≈ 63.3 km/h

4. Same: least speed

ANSWER ≈ 56.9 km/h

5. A square of side 5.0 cm (1 d.p.): area bounds

ANSWER 24.5025 to 25.5025

6. A cube of edge 4 cm (nearest cm): volume bounds

ANSWER 42.875 to 91.125

7. To how many figures can that volume be quoted?

ANSWER none — not even the first digit agrees

Hard · stretch and reason

7 PROBLEMS

1. A circle of radius 6.0 cm (1 d.p.): area bounds

ANSWER ≈ 111.2 to ≈ 115.0

2. A cylinder $r = 3$, $h = 10$ (nearest 1): volume bounds

ANSWER ≈ 205.6 to ≈ 405.7

3. A density is $\frac{m}{V}$ with $m = 250$ g (nearest 10) and $V = 50$ cm³ (nearest 5). Greatest density

ANSWER ≈ 5.42 g/cm³

4. Same: least density

ANSWER ≈ 4.62 g/cm³

5. A rectangle has area 40 cm² and length 8 cm, both to the nearest unit. Greatest width

ANSWER ≈ 5.4 cm

6. Explain why bounds widen faster for a volume than for a length

ANSWER Three uncertainties multiply

7. A pipe is 2.0 m long to 1 d.p.; how many fit in 50 m to the nearest metre?

ANSWER at least 24, at most 25

07 Homework

Homework — 4 tasks

Always state both bounds, and say which extreme you used and why.

- Write the bounds of 15 cm (nearest cm), 3.7 m (1 d.p.) and 600 g (nearest 100 g).
- A cuboid measures $4 \times 6 \times 9$ cm, each to the nearest cm. Find the bounds of its volume.
- A car travels 80 km (nearest 10 km) in 1.5 h (nearest 0.1 h). Find the greatest and least possible speeds.

4. Explain, with the cuboid of task 2, why the answer cannot honestly be quoted to three figures.

The concept of volume, and the volume of a parallelepiped

What a volume is before any formula — three properties, and the cuboid they force.

LESSON 25

1 × 40 MIN

GEOMETRY 11, §12

IGCSE E13.1

LESSON CLOCK

10'

12'

12'

8'

3'

What volume is	10 min
The cuboid	12 min
The parallelepiped	12 min
Units	8 min
Homework	3 min

LEARNING OBJECTIVES

- State the three defining properties of volume.
- Derive $V = abc$ for a cuboid from those properties.
- Compute the volume of a parallelepiped.
- Convert between units of volume.

TEXTBOOK

National	pp. 123–128
Cambridge	Core & Extended, pp. 301–304
Workbook	Exercise 12.1

01

Explanation

The three properties

VOLUME IS DEFINED BY WHAT IT DOES

1 Congruent solids have equal volumes. **2** If a solid is cut into pieces, the volumes add. **3** The unit cube has volume 1.

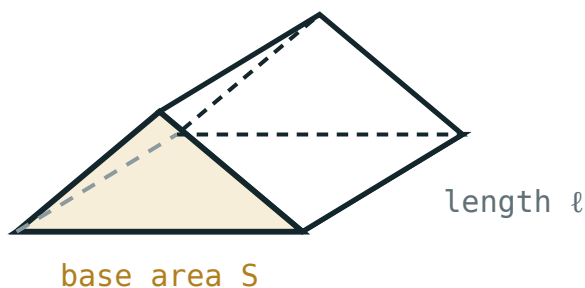
Nothing else is assumed. Every formula in this chapter is a consequence of these three statements, and the first formula follows almost immediately.

The cuboid

A cuboid $a \times b \times c$ with whole-number edges can be filled with exactly abc unit cubes: ab in the bottom layer, and c layers. By properties 2 and 3 its volume is abc .

$$V = abc = B \cdot h$$

For fractional and then irrational edges the same result follows by squeezing between cuboids with rational edges. The formula, once proved for whole numbers, holds for all.



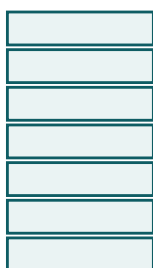
$$V = S \cdot l$$

The bottom layer holds ab cubes; there are c layers.

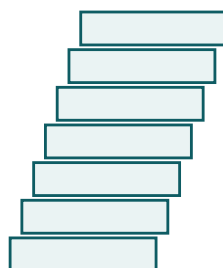
The parallelepiped

An oblique parallelepiped has the same volume as the right one on the same base with the same height. The proof is Cavalieri's principle: every horizontal slice of the two solids has the same area, so the volumes agree.

same base, same height, same volume



right prism



oblique prism

every horizontal slice has the same area

Slide the layers sideways. Nothing is added or removed, so the volume is unchanged.

$V = B \cdot h$ for every parallelepiped, right or oblique

h IS THE PERPENDICULAR HEIGHT

Not the lateral edge. In an oblique solid the lateral edge is longer than the height, and using it inflates the answer.

Units

LENGTH	VOLUME	FACTOR
$1\text{ cm} = 10\text{ mm}$	$1\text{ cm}^3 = 1000\text{ mm}^3$	10^3
$1\text{ m} = 100\text{ cm}$	$1\text{ m}^3 = 1\,000\,000\text{ cm}^3$	100^3
—	$1\text{ litre} = 1000\text{ cm}^3$	—
—	$1\text{ m}^3 = 1000\text{ litres}$	—

CUBE THE LENGTH FACTOR
 1 m^3 is not 100 cm^3 but $1\,000\,000$. This is the k^3 rule of Grade 11 similarity, applied to a change of unit.

02 Worked examples

EXAMPLE 1 A cuboid is $12 \times 8 \times 5\text{ cm}$. Find its volume in cm^3 and in litres.

1 $V = 480\text{ cm}^3$.

2 $1\text{ litre} = 1000\text{ cm}^3$.

3 0.48 litres .

ANSWER
 $480\text{ cm}^3 = 0.48\text{ litres}$

EXAMPLE 2 An oblique parallelepiped has base area 30 cm^2 and lateral edge 10 cm inclined at 60° to the base. Find its volume.

① Height $h = 10 \sin 60^\circ = 5\sqrt{3} \approx 8.66$.

Not 10 .

② $V = 30 \times 8.66$

③ ≈ 259.8

ANSWER

$$150\sqrt{3} \approx 260 \text{ cm}^3$$

EXAMPLE 3 A tank holds 2.4 m^3 . How many litres is that?

① $1 \text{ m}^3 = 1000 \text{ litres}$.

② 2.4×1000

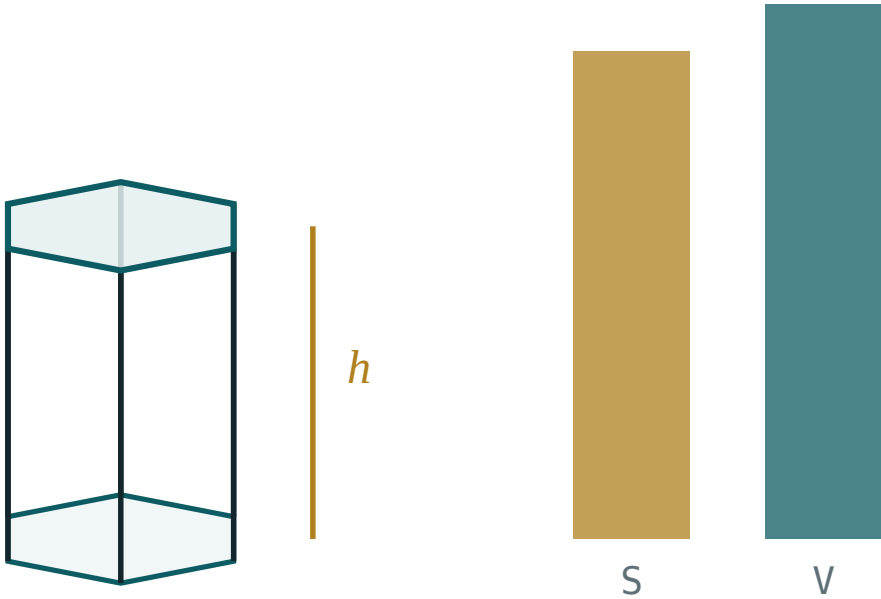
ANSWER

2400 litres

03 Interactive model

Fill a box with unit cubes, one layer at a time, and count aloud.

Base area times height



base area B 41.6

perimeter P 24

total S 227.1

volume V 249.4

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Volume	Hajm	Объём
Unit cube	Birlik kub	Единичный куб
Additivity	Additivlik	Аддитивность
Congruent solids	Teng jismlar	Равные тела
Cubic unit	Kub birlik	Кубическая единица
Parallelepiped	Parallelepiped	Параллелепипед
Base area	Asos yuzasi	Площадь основания
Height of a solid	Jism balandligi	Высота тела
Capacity	Sig'im	Вместимость

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The volume of a unit cube is:

Volume of a cuboid $a \times b \times c$:

1 m³ in cm³:

1 litre in cm^3 :

100

1000

10 000

1

For an oblique solid $V = Bh$ where h is:

the lateral edge

the perpendicular height

the diagonal

the base edge

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Volume of a $3 \times 4 \times 5$ cuboid

ANSWER 60

2. Volume of a cube of edge 6

ANSWER 216

3. 1 m^3 in litres

ANSWER 1000

4. 2500 cm^3 in litres

ANSWER 2.5

5. 1 cm^3 in mm^3

ANSWER 1000

6. A cube of volume 125: its edge

ANSWER 5

7. Volume with $B = 20$, $h = 7$

ANSWER 140

Medium · the standard question

7 PROBLEMS

1. Cuboid $12 \times 8 \times 5$ in litres

ANSWER 0.48

2. Oblique parallelepiped, $B = 30$, edge 10 at 60°

ANSWER $150\sqrt{3} \approx 260$

3. A tank of 2.4 m^3 in litres

ANSWER 2400

4. A cuboid of volume 720 with base 12×10 : its height

ANSWER 6

5. A cube of surface area 216: its volume

ANSWER 216

6. Convert 0.75 m^3 to cm^3

ANSWER 750 000

7. A regular hexagonal prism, base edge 4, height 10: volume

ANSWER $240\sqrt{3} \approx 416$

Hard · stretch and reason

7 PROBLEMS

1. A parallelepiped has edges 6, 8, 10 with the 10-edge at 45° to the base. Volume

ANSWER $240\sqrt{2} \approx 339$

2. A cuboid of volume V is scaled by k . New volume

ANSWER k^3V

3. A cube of volume 64 cm^3 : its surface area

ANSWER 96 cm^2

4. A tank $2 \text{ m} \times 1.5 \text{ m} \times 0.8 \text{ m}$ holds how many litres?

ANSWER 2400

5. A brick is $24 \times 12 \times 6 \text{ cm}$. How many in a 1 m^3 wall?

ANSWER 578 (allowing no mortar)

6. Prove $V = abc$ for whole-number edges from the three properties

ANSWER Fill with abc unit cubes and add

7. Why does the same formula hold for irrational edges?

ANSWER Squeeze between rational cuboids

07 Homework

Homework — 4 tasks

Always state the unit, and convert with the cube of the length factor.

- Find the volume of a $15 \times 9 \times 4 \text{ cm}$ cuboid in cm^3 and in litres.
- An oblique parallelepiped has base area 48 cm^2 and lateral edge 12 cm inclined at 30° . Find its volume.
- Convert 3.6 m^3 to litres and to cm^3 .
- Explain in three sentences why 1 m^3 is a million cm^3 and not a hundred.

The volume of a prism

Base area times height, for every prism — and the dissection that proves it.

LESSON 26–27

2 × 40 MIN

GEOMETRY 11, §13

IGCSE E13.X

LESSON CLOCK

14'

20'

20'

22'

16'

8'

From the cuboid to any prism	14 min
Applying the formula	20 min
Working backwards	20 min
Compound solids	22 min
Practice	16 min
Homework	8 min

LEARNING OBJECTIVES

- Derive $V = Bh$ for a triangular prism from the parallelepiped.
- Compute the volume of any prism.
- Solve for a missing dimension from a given volume.
- Handle compound solids made of prisms.

TEXTBOOK

National	pp. 129–140
Cambridge	Core & Extended, pp. 305–310
Workbook	Exercise 13.1

01

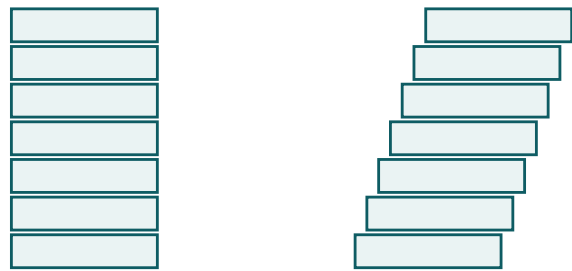
Explanation

From the cuboid to any prism

- Step 1.** A parallelepiped is a cuboid with the layers slid sideways, so $V = Bh$ by Cavalieri.
- Step 2.** A triangular prism is half a parallelepiped, cut along a diagonal plane. Both halves are congruent, so each has volume $\frac{1}{2}B_p h = B h$, where B is now the triangle's area.
- Step 3.** Any polygonal base splits into triangles; the prism splits into triangular prisms of the same height, and the volumes add.

$$V = B \cdot h \text{ for every prism}$$

same base, same height, same volume



right prism oblique prism
every horizontal slice has the same area

Cavalieri's principle: equal slices at every height mean equal volumes.

THE IDEA BEHIND EVERY VOLUME FORMULA

A solid of uniform cross-section has volume = area of cross-section × length. The cylinder of the next lesson is the same statement with a circle instead of a polygon.

Applying it

PRISM	B	V
cuboid $a \times b$, height h	ab	abh
right triangular, legs p, q	$\frac{1}{2}pq$	$\frac{1}{2}pqh$
regular triangular, edge a	$\frac{\sqrt{3}}{4}a^2$	$\frac{\sqrt{3}}{4}a^2h$
regular hexagonal, edge a	$\frac{3\sqrt{3}}{2}a^2$	$\frac{3\sqrt{3}}{2}a^2h$
trapezium base p, q , depth d	$\frac{p+q}{2}d$	$\frac{p+q}{2}dh$

Compound solids

Real objects — an L-shaped room, a channel, a house with a roof — are prisms with a compound cross-section. Find the area of the whole cross-section first, then multiply once.

Example. A shed is a cuboid 6×4 m and 2.5 m to the eaves, with a triangular roof 1.5 m high. Cross-section: rectangle $4 \times 2.5 = 10$ plus triangle $\frac{1}{2} \times 4 \times 1.5 = 3$, total 13 m^2 . Volume $13 \times 6 = 78 \text{ m}^3$.

MULTIPLY ONCE, AT THE END

Computing each part's volume separately works too, but doubles the arithmetic and the chances of error. Add the areas first.

02 Worked examples

EXAMPLE 1 A regular hexagonal prism has base edge 5 cm and height 12 cm. Find its volume.

$$\textcircled{1} \quad B = \frac{3\sqrt{3}}{2} \times 25 \approx 64.95$$

$$\textcircled{2} \quad V = 64.95 \times 12$$

$$\textcircled{3} \quad \approx 779.4$$

ANSWER

$$150\sqrt{3} \times \frac{12}{2} \text{ — i.e. } \approx 779 \text{ cm}^3$$

EXAMPLE 2 A trough has a trapezium cross-section with parallel sides 40 cm and 60 cm, depth 30 cm, and is 2 m long. Find its capacity in litres.

$$\textcircled{1} \quad B = \frac{40 + 60}{2} \times 30 = 1500 \text{ cm}^2.$$

$$\textcircled{2} \quad V = 1500 \times 200 = 300\,000 \text{ cm}^3.$$

$$\textcircled{3} \quad = 300 \text{ litres.}$$

ANSWER

300 litres

EXAMPLE 3 A prism has volume 540 cm³ and a triangular base with legs 6 and 9. Find its height.

$$\textcircled{1} \quad B = \frac{1}{2} \times 6 \times 9 = 27$$

$$\textcircled{2} \quad 27h = 540$$

$$\textcircled{3} \quad h = 20$$

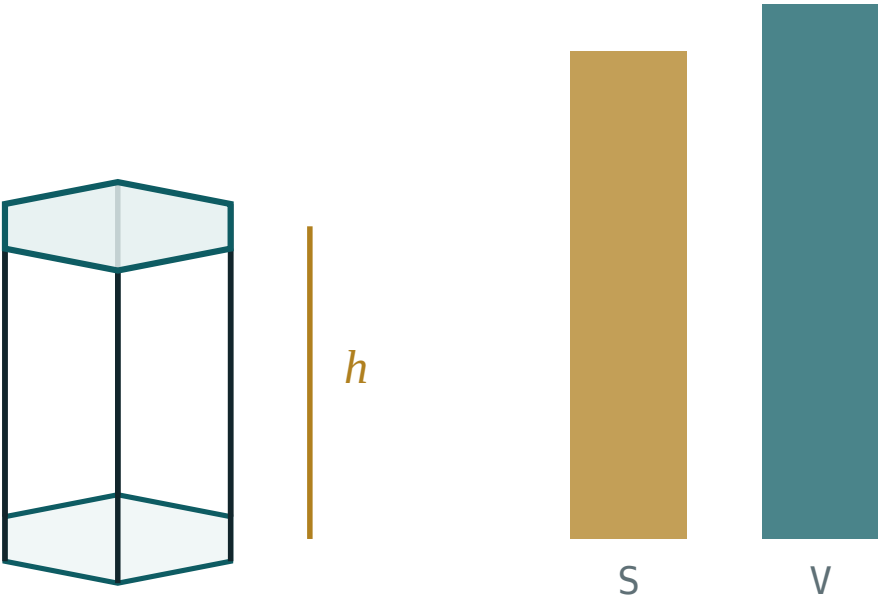
ANSWER

20 cm

03 Interactive model

Cut a parallelepiped of plasticine along a diagonal plane and compare the two halves.

The volume of a prism



base area B 41.6

perimeter P 24

total S 227.1

volume V 249.4

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Volume of a prism	Prizma hajmi	Объём призмы
Base area	Asos yuzasi	Площадь основания
Height	Balandlik	Высота
Dissection	Bo'lish	Разбиение
Cavalieri's principle	Kavaleri prinsipi	Принцип Кавальери
Compound solid	Murakkab jism	Составное тело
Cross-section	Ko'ndalang kesim	Поперечное сечение
Uniform cross-section	O'zgarmas kesim	Постоянное сечение
Capacity	Sig'im	Вместимость

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The volume of any prism is:

Ph

Bh

$2B + Ph$

$\frac{1}{3} Bh$

A triangular prism is:

a third of a cuboid

half a parallelepiped

a quarter of a cube

unrelated

Cavalieri's principle says:

equal bases give equal volumes

equal slice areas at every height give equal volumes

volumes always add

nothing about volume

A prism of $B = 27$ and $V = 540$ has height:

15

20

27

30

For a compound cross-section you should:

multiply each part

add the areas then multiply once

take the largest

average them

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Volume with $B = 12$, $h = 5$

ANSWER 60

2. Cuboid $4 \times 5 \times 6$

ANSWER 120

3. Right triangular prism, legs 3 and 4, height 10

ANSWER 60

4. Cube of edge 8

ANSWER 512

5. Prism $B = 25$, $h = 4$

ANSWER 100

6. A prism of $V = 100$, $B = 20$: the height

ANSWER 5

7. Regular triangular prism, edge 4, height 6

ANSWER $24\sqrt{3}$

Medium · the standard question

7 PROBLEMS

1. Regular hexagonal prism, edge 5, height 12

ANSWER $\approx 779 \text{ cm}^3$

2. Trough, trapezium 40/60 by 30 cm, 2 m long: litres

ANSWER 300

3. Prism $V = 540$, triangular base legs 6 and 9: height

ANSWER 20

4. Regular hexagonal prism, edge 4, height 10

ANSWER $240\sqrt{3} \approx 416$

5. Shed: rectangle 4×2.5 plus roof triangle height 1.5, length 6 m

ANSWER 78 m^3

6. An L-shaped cross-section of area 34 cm^2 , length 15 cm

ANSWER 510 cm^3

7. A cuboid of volume 600 with base 10×12 : height

ANSWER 5

Hard · stretch and reason

7 PROBLEMS

1. A prism has a rhombus base with diagonals 10 and 24, height 15. Volume

ANSWER 1800

2. A regular hexagonal prism of volume $648\sqrt{3}$ and edge 6: its height

ANSWER 12

3. A channel of trapezium section 1.2 m and 2 m wide, 0.8 m deep, 30 m long: capacity in m^3

ANSWER 38.4

4. Two similar prisms of volumes 54 and 128: the surface-area ratio

ANSWER 9 : 16

5. A cube is cut by a plane through A , C , B_1 . Find the volume of the smaller piece for edge 6

ANSWER 36

6. A prism's height doubles and its base edges halve. What happens to the volume?

ANSWER It halves

7. Prove $V = Bh$ for a triangular prism

ANSWER It is half a parallelepiped on twice the base

07 Homework

Homework — 5 tasks

Add the cross-sectional areas before multiplying by the length.

1. A regular hexagonal prism has base edge 6 cm and height 14 cm. Find its volume.
2. A trough has a trapezium section with parallel sides 30 cm and 50 cm and depth 25 cm, and is 3 m long. Find its capacity in litres.
3. A prism has volume 420 cm^3 and a triangular base with legs 7 and 8. Find its height.

4. A shed has a rectangular cross-section 5×3 m with a triangular roof 2 m high, and is 8 m long. Find its volume.
5. Prove that a triangular prism has volume Bh , starting from the parallelepiped.

The cylinder — surface area and volume

A prism whose base is a circle: the same two formulas, with π in them.

LESSON 28–29

2 × 40 MIN

GEOMETRY 11, §14

IGCSE E13.X

LESSON CLOCK

12'

22'

22'

22'

20'

8'

A prism with a circular base	12 min
The net and the area	22 min
The volume	22 min
Working backwards	22 min
Compound solids	20 min
Homework	8 min

LEARNING OBJECTIVES

- Derive the surface area of a cylinder from its net.
- Compute the volume of a cylinder.
- Solve for a missing radius or height.
- Handle hollow cylinders and compound solids.

TEXTBOOK

National	pp. 141–152
Cambridge	Core & Extended, pp. 311–318
Workbook	Exercise 14.1

01

Explanation

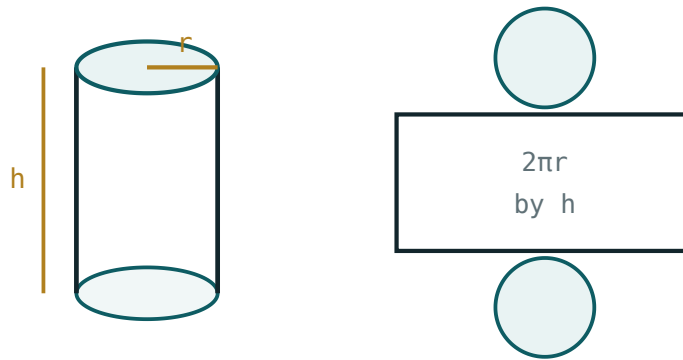
A prism with a circular base

A **cylinder** is the solid swept out when a rectangle turns about one of its sides. Its two bases are congruent circles in parallel planes, and every cross-section parallel to the base is the same circle.

So the principle of the last lesson applies unchanged: **uniform cross-section means $V = Bh$** , with $B = \pi r^2$.

$$V = \pi r^2 h$$

The net and the surface area



$$S = 2\pi r^2 + 2\pi rh - \text{two circles and one rectangle}$$

Cut down one side and unroll: two circles and a rectangle $2\pi r$ wide.

The curved surface unrolls into a rectangle whose width is the circumference of the base and whose height is the cylinder's height:

$$S_{\text{curved}} = 2\pi rh \quad S_{\text{total}} = 2\pi r^2 + 2\pi rh = 2\pi r(r + h)$$

CURVED, OR TOTAL?

A tin has two ends. A pipe has none. A cup has one. Read the object before choosing the formula — the question always describes it precisely enough to decide.

Working backwards

Given a volume and one dimension, the other follows:

$$r = \sqrt{\frac{V}{\pi h}} \quad h = \frac{V}{\pi r^2}$$

Example. A cylindrical tin holds 500 ml and is 12 cm tall. Its radius is $\sqrt{\frac{500}{12\pi}} \approx 3.64$ cm.

Given a **surface area** and one dimension, the equation is quadratic in r — solve it with the formula and reject the negative root.

Hollow and compound solids

A pipe of outer radius R and inner radius r is the difference of two cylinders:

$$V = \pi(R^2 - r^2)h$$

Do **not** compute $\pi(R - r)^2h$ — the cross-section is an annulus, whose area is a difference of two circles, not the circle on the difference of the radii.

Compound solids. A silo is a cylinder with a hemisphere on top; a pencil is a hexagonal prism with a cone. Split, compute each, add — and remember that the joining faces are internal and contribute nothing to the surface area.

THE AXIAL SECTION

Cutting a cylinder through its axis gives a rectangle $2r$ by h . A cylinder whose axial section is a square has $h = 2r$ — the shape that minimises the surface area for a fixed volume, as Lesson 23–25 of Quarter I showed.

02 Worked examples

EXAMPLE 1 A cylinder has radius 5 cm and height 12 cm. Find its volume and total surface area.

$$\begin{aligned} 1 \quad V &= \pi \times 25 \times 12 = 300\pi \\ &\approx 942.5 \text{ cm}^3 \end{aligned}$$

$$2 \quad S = 2\pi(25) + 2\pi(5)(12)$$

$$\begin{aligned} 3 \quad &= 50\pi + 120\pi = 170\pi \\ &\approx 534.1 \text{ cm}^2 \end{aligned}$$

ANSWER

$$300\pi \approx 942 \text{ cm}^3; 170\pi \approx 534 \text{ cm}^2$$

EXAMPLE 2 A cylindrical tin holds 500 ml and is 12 cm tall. Find its radius.

$$\begin{aligned} 1 \quad 500 &= \pi r^2 \times 12 \\ 1 \text{ ml} &= 1 \text{ cm}^3. \end{aligned}$$

$$2 \quad r^2 = \frac{500}{12\pi} \approx 13.26$$

$$3 \quad r \approx 3.64$$

ANSWER

$\approx 3.64 \text{ cm}$

EXAMPLE 3 A pipe is 2 m long with outer radius 8 cm and inner radius 6 cm. Find the volume of metal.

$$\begin{aligned} 1 \quad V &= \pi(64 - 36) \times 200 \\ \text{Length in cm.} \end{aligned}$$

$$2 \quad = \pi \times 28 \times 200$$

$$3 \quad = 5600\pi \approx 17\,593$$

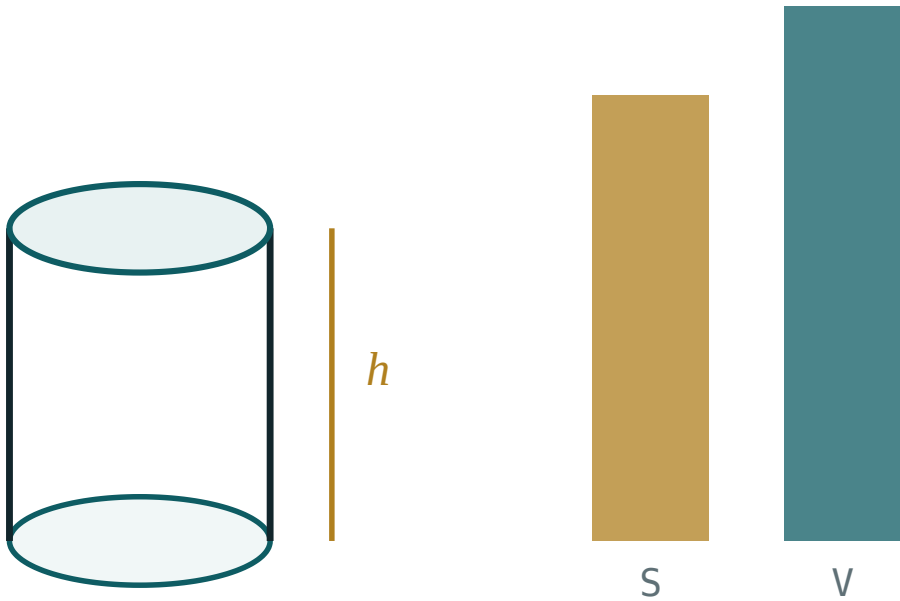
ANSWER

$5600\pi \approx 17\,600 \text{ cm}^3$

03 Interactive model

Unroll a tin label and measure the rectangle — its width is the circumference.

The cylinder



base area B **50.3**

perimeter P **25.1**

total S **251.3**

volume V **301.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Cylinder	Silindr	Цилиндр
Axis of a cylinder	Silindr o'qi	Ось цилиндра
Base circle	Asos aylanasi	Основание
Curved surface	Yon sirt	Боковая поверхность
Net of a cylinder	Silindr yoymasi	Развёртка цилиндра
Axial section	O'qli kesim	Осевое сечение
Hollow cylinder (pipe)	Bo'sh silindr	Полый цилиндр
Capacity	Sig'im	Вместимость
Solid of revolution	Aylanish jismi	Тело вращения

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The volume of a cylinder is:

$$2\pi rh$$

$$\pi r^2 h$$

$$\pi rh$$

$$\frac{1}{3} \pi r^2 h$$

The curved surface area is:

$$\pi r^2$$

$$2\pi rh$$

$$2\pi r^2$$

$$\pi r^2 h$$

The total surface area is:

$$2\pi rh$$

$$2\pi r(r + h)$$

$$\pi r(r + h)$$

$$\pi r^2 h$$

A pipe's cross-section area is:

$$\pi(R - r)^2$$

$$\pi(R^2 - r^2)$$

$$\pi R^2$$

$$2\pi R$$

A cylinder with a square axial section has:

$$h = r$$

$$h = 2r$$

$$h = 4r$$

$$r = 2h$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Volume, $r = 3$, $h = 10$

ANSWER $90\pi \approx 283$

2. Volume, $r = 2$, $h = 7$

ANSWER $28\pi \approx 88$

3. Curved area, $r = 5$, $h = 4$

ANSWER 40π

4. Total area, $r = 1$, $h = 3$

ANSWER 8π

5. A tin of $r = 4$, $h = 10$: volume in litres

ANSWER ≈ 0.503

6. Volume, $r = 10$, $h = 10$

ANSWER 1000π

7. Curved area, $r = 7$, $h = 2$

ANSWER 28π

Medium · the standard question

7 PROBLEMS

1. $r = 5, h = 12$: volume and total area

ANSWER $300\pi; 170\pi$

2. A tin holds 500 ml and is 12 cm tall: its radius

ANSWER ≈ 3.64 cm

3. A cylinder of volume 1000π and radius 10: its height

ANSWER 10

4. Pipe, 2 m long, $R = 8, r = 6$: metal volume

ANSWER $5600\pi \approx 17\,600$ cm³

5. A cylinder with a square axial section and $r = 6$: its volume

ANSWER 432π

6. A cup (no lid), $r = 4, h = 9$: surface area

ANSWER 88π

7. A cylinder of total area 100π and $r = 5$: its height

ANSWER 5

Hard · stretch and reason

7 PROBLEMS

1. A closed cylinder of volume 1000 cm^3 with least surface area: find r

ANSWER $\approx 5.42 \text{ cm}$

2. A silo: cylinder $r = 3$, $h = 10$ with a hemisphere on top. Volume

ANSWER $108\pi \approx 339$

3. Same silo: external surface area (no base)

ANSWER 78π

4. A cylinder is inscribed in a cube of edge 10. Its volume

ANSWER 250π

5. A cylinder of height 15 has curved area 120π . Find r and V

ANSWER $r = 4$, $V = 240\pi$

6. A pipe of length L , outer R , wall thickness t : metal volume

ANSWER $\pi L(2Rt - t^2)$

7. Two similar cylinders of volumes 27 and 64: the area ratio

ANSWER $9 : 16$

07 Homework

Homework — 5 tasks

Say whether the solid is closed, open or hollow before choosing the formula.

1. A cylinder has radius 6 cm and height 15 cm. Find its volume and total surface area.
2. A cylindrical can holds 750 ml and has radius 5 cm. Find its height.
3. A pipe is 3 m long with outer radius 10 cm and inner radius 8 cm. Find the volume of metal in cm^3 .
4. A cylinder has a square axial section and radius 7 cm. Find its volume and total surface area.

5. A silo is a cylinder of radius 4 m and height 12 m with a hemispherical top. Find its capacity in m^3 .

Units of volume and capacity, and converting between them

Cambridge insert: the cube of the length factor, and the litre that sits between the two systems.

LESSON 30

1 × 40 MIN

GEOMETRY 11, §14 (EXTENSION)

IGCSE E13.1–13.4

LESSON CLOCK

10'

12'

12'

8'

3'

Cubing the factor	10 min
Litres	12 min
Choosing a unit	12 min
Density	8 min
Homework	3 min

LEARNING OBJECTIVES

- Convert between mm^3 , cm^3 and m^3 .
- Convert between millilitres, litres and cubic units.
- Choose an appropriate unit for a given object.
- Use density to link volume and mass.

TEXTBOOK

National	pp. 153–156
Cambridge	Core & Extended, pp. 40–44
Workbook	IGCSE Exercise 13.1

01

Explanation

Cube the length factor

THE ONE RULE

If lengths convert by k , volumes convert by k^3 . $1\text{ m} = 100\text{ cm}$, so $1\text{ m}^3 = 100^3 = 1\,000\,000\text{ cm}^3$.

CONVERSION	LENGTH FACTOR	VOLUME FACTOR
cm → mm	10	1000
m → cm	100	1 000 000
km → m	1000	10 ⁹
m → mm	1000	10 ⁹

THE COMMONEST CONVERSION ERROR IN THE SYLLABUS

Writing $1\text{ m}^3 = 100\text{ cm}^3$. Areas need k^2 and volumes k^3 ; only lengths use k itself.

Litres

$$1\text{ ml} = 1\text{ cm}^3 \quad 1\text{ litre} = 1000\text{ cm}^3 \quad 1\text{ m}^3 = 1000\text{ litres}$$

The litre is defined as a cubic decimetre — a cube of side 10 cm . That single fact generates every line above, and it is why a 1 m^3 tank holds exactly a tonne of water.

OBJECT	TYPICAL CAPACITY
a teaspoon	5 ml
a can of drink	330 ml
a bucket	10 litres
a bath	150 litres
a small water tank	$1\text{ m}^3 = 1000\text{ litres}$

Choosing a unit, and density

An answer of 0.0000034 m^3 or $34\,000\,000\text{ mm}^3$ is correct and useless. Choose the unit that makes the number sit between about 0.1 and 1000 .

$$\text{density } \rho = \frac{m}{V} \text{ so } m = \rho V$$

Water has $\rho = 1\text{ g/cm}^3$, which is why 1 litre of water weighs 1 kg and 1 m^3 weighs 1 tonne . Steel is about 7.8 g/cm^3 , pine about 0.5 .

Example. A steel bar is a cylinder $r = 2\text{ cm}$, $h = 50\text{ cm}$. Its volume is $200\pi \approx 628\text{ cm}^3$, so its mass is about $628 \times 7.8 \approx 4900\text{ g} \approx 4.9\text{ kg}$.

02 Worked examples**EXAMPLE 1** Convert 2.5 m^3 to cm^3 and to litres.

① $1 \text{ m}^3 = 1\,000\,000 \text{ cm}^3$.

② $2\,500\,000 \text{ cm}^3$.

③ $\div 1000$ gives 2500 litres.

ANSWER

$$2\,500\,000 \text{ cm}^3 = 2500 \text{ litres}$$

EXAMPLE 2 A tank is $1.2 \times 0.8 \times 0.5 \text{ m}$. How many litres does it hold?

① $V = 0.48 \text{ m}^3$.

② $\times 1000$

③ $= 480 \text{ litres}$.

ANSWER

$$480 \text{ litres}$$

EXAMPLE 3 A steel cylinder has $r = 2$ cm and $h = 50$ cm. Find its mass at 7.8 g/cm³.

① $V = \pi \times 4 \times 50 = 200\pi \approx 628.3$

② $m = 628.3 \times 7.8$

③ ≈ 4901 g.

ANSWER

≈ 4.9 kg

03 Interactive model

Show a litre bottle beside a 10 cm cube — they hold the same.

Converting volume

1 m³ in cm³:

100

10 000

1 000 000

1000

1 cm³ in mm³:

10

100

1000

10 000

1 litre in cm³:

100

1000

10 000

1

1 m³ in litres:

100

1000

10 000

1 000 000

0.5 m³ in litres:

50

500

5000

5

1 litre of water weighs:

100 g

1 kg

10 kg

1 g

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Cubic centimetre	Kub santimetr	Кубический сантиметр
Cubic metre	Kub metr	Кубический метр
Millilitre	Millilitr	Миллилитр
Litre	Litr	Литр
Capacity	Sig'im	Вместимость
Conversion factor	O'tkazish koeffitsienti	Коэффициент перевода
Density	Zichlik	Плотность
Mass	Massa	Масса
Appropriate unit	Mos birlik	Подходящая единица

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Volumes convert by:

k

k^2

k^3

$3k$

1 ml equals:

1 mm³

1 cm³

10 cm³

1 litre

3 m³ in litres:

300

3000

30 000

3

Density is:

$$mV$$

$$\frac{m}{V}$$

$$\frac{V}{m}$$

$$m + V$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 1 m^3 in cm^3

ANSWER $1\,000\,000$

2. 1 cm^3 in mm^3

ANSWER 1000

3. 2 litres in cm^3

ANSWER 2000

4. 5000 cm^3 in litres

ANSWER 5

5. 0.25 m^3 in litres

ANSWER 250

6. 750 ml in cm^3

ANSWER 750

7. 1 litre of water in kg

ANSWER 1

Medium · the standard question

7 PROBLEMS

1. 2.5 m^3 in cm^3 and litres

ANSWER $2\,500\,000 \text{ cm}^3$, 2500 L

2. Tank $1.2 \times 0.8 \times 0.5 \text{ m}$ in litres

ANSWER 480

3. A cylinder $r = 5$, $h = 20 \text{ cm}$ in litres

ANSWER ≈ 1.57

4. A steel cylinder $r = 2$, $h = 50 \text{ cm}$ at 7.8 g/cm^3 : mass

ANSWER $\approx 4.9 \text{ kg}$

5. $3\,500\,000 \text{ mm}^3$ in litres

ANSWER 3.5

6. A bath of 150 litres in m^3

ANSWER 0.15

7. A block of pine $20 \times 10 \times 5 \text{ cm}$ at 0.5 g/cm^3 : mass

ANSWER 500 g

Hard · stretch and reason

7 PROBLEMS

1. A pool $25 \times 10 \times 1.8$ m: capacity in litres

ANSWER 450 000

2. A pipe of length 5 m, $R = 6$, $r = 5$ cm: metal mass at 7.8 g/cm^3

ANSWER $\approx 135 \text{ kg}$

3. A raindrop of radius 2 mm: volume in ml

ANSWER ≈ 0.0335

4. A tonne of water occupies how many m^3 ?

ANSWER 1

5. A gold bar $10 \times 5 \times 3$ cm at 19.3 g/cm^3 : mass

ANSWER $\approx 2.9 \text{ kg}$

6. Explain why 1 m^3 is not 100 cm^3

ANSWER Volumes need the cube of the length factor

7. A tank holds 2400 litres and is 2×1.5 m in base. Find its depth

ANSWER 0.8 m

07 Homework

Homework — 4 tasks

Choose the unit that keeps the number readable, and say why.

- Convert 4.2 m^3 to cm^3 and to litres, and 6500 mm^3 to cm^3 .
- A tank measures $2 \times 1.5 \times 0.9$ m. Find its capacity in litres.
- A steel pipe 4 m long has outer radius 7 cm and inner radius 6 cm. Find the mass of metal at 7.8 g/cm^3 .

4. Explain in three sentences why volumes convert with the cube of the length factor, with one worked example.

Control work 2, and the quarter review

Prisms and cylinders — area, volume and bounds — in one paper, then the map.

LESSON 31–32

2 × 40 MIN

GEOMETRY 11, NAZORAT ISHI 2

IGCSE E13 REVIEW

LESSON CLOCK

3

42'

12'

20'

10'

3

Instructions	3 min
The paper	42 min
Answers	12 min
Rewrite	20 min
Concept map	10 min
Targets	3 min

LEARNING OBJECTIVES

- Apply every formula of Quarter II in one assessment.
- Choose between curved, total and open-solid surface areas.
- Classify each lost mark and rewrite the solution.
- Build a concept map of mensuration so far.

TEXTBOOK

National	pp. 157–160
Cambridge	Core & Extended, pp. 319–322
Workbook	Control paper G2

01

Explanation

The paper — 30 marks, 42 minutes

Q	TASK	MARKS	FROM
1	Space diagonal of a $4 \times 6 \times 12$ cuboid, and the angle it makes with the base	5	L21–22
2	Total surface area of a regular hexagonal prism, edge 5, height 11	5	L23
3	Volume of a trough with trapezium section 30/50 cm, depth 20 cm, 2.5 m long, in litres	5	L26–27
4	A cylinder holds 800 ml and is 10 cm tall: find r and the curved surface area	5	L28–29
5	A pipe 2 m long, $R = 9$, $r = 7$ cm: metal volume, then mass at 7.8 g/cm^3	5	L28–30
6	A cuboid measures $5 \times 7 \times 9$ cm to the nearest cm. Find the bounds of its volume	5	L24

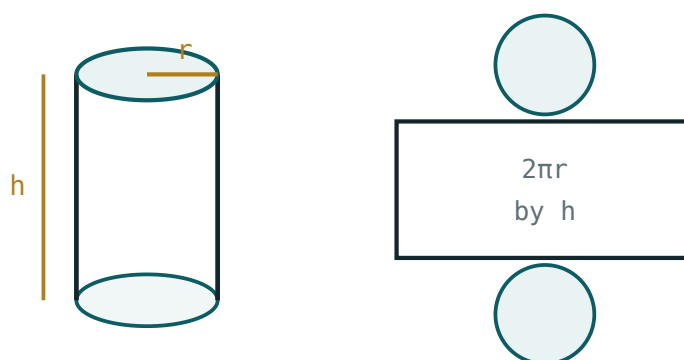
UNITS AND FORMULA CHOICE

Q3 and Q5 each carry a mark for the conversion alone; Q4 carries one for choosing **curved** rather than total. Five of the thirty marks are for reading the question.

The concept map

Six boxes, links as sentences:

- **polyhedral angle** → **only five regular solids** — “the angles must total under 360° ”
- **prism** → **uniform cross-section** — “ $V = Bh$, always”
- **net** → **surface area** — “ $S = 2B + Ph$ ”
- **cylinder** → **the same two formulas** — “with $B = \pi r^2$ and $P = 2\pi r$ ”
- **Cavalieri** → **oblique = right** — “equal slices, equal volume”
- **measurement** → **bounds** — “three lengths, three uncertainties, one wide answer”



$$S = 2\pi r^2 + 2\pi rh - \text{two circles and one rectangle}$$

The picture behind two of the six boxes.

Looking forward

Quarter III is the pyramid, the cone and the sphere — the solids whose volume formulas carry a $\frac{1}{3}$ or a $\frac{4}{3}$. Every one is proved by comparison with the prism and cylinder of this quarter, so those two formulas must be secure.

ONE HABIT TO CARRY FORWARD

Write the base area on its own line before anything else. Every volume in the next quarter starts from B , and every error starts from having computed it wrong.

02 Worked examples

EXAMPLE 1 Model answer, Q3: trough, trapezium 30/50 cm, depth 20 cm, 2.5 m long.

$$1 \quad B = \frac{30 + 50}{2} \times 20 = 800 \text{ cm}^2.$$

2 Length 250 cm.
Convert first.

$$3 \quad V = 800 \times 250 = 200\,000 \text{ cm}^3.$$

$$4 \quad = 200 \text{ litres.}$$

ANSWER

200 litres

EXAMPLE 2 Model answer, Q4: a cylinder holds 800 ml and is 10 cm tall.

$$1 \quad 800 = \pi r^2 \times 10$$

$$2 \quad r^2 = \frac{80}{\pi} \approx 25.46$$

$$3 \quad r \approx 5.05 \text{ cm.}$$

$$4 \quad S_{\text{curved}} = 2\pi rh \approx 317 \text{ cm}^2.$$

Curved, not total.

ANSWER

$r \approx 5.05 \text{ cm}; \approx 317 \text{ cm}^2$

EXAMPLE 3 Model answer, Q6: bounds for a $5 \times 7 \times 9$ cuboid to the nearest cm.

- ① 4.5–5.5, 6.5–7.5, 8.5–9.5.
- ② $V_{\min} = 4.5 \times 6.5 \times 8.5 = 248.625$
- ③ $V_{\max} = 5.5 \times 7.5 \times 9.5 = 391.875$
- ④ Stated volume 315.
The range is nearly 150.

ANSWER248.6 to 391.9 cm³**03** Interactive model

Work Q3 and Q4 on the board, converting units before any arithmetic.

The quarter in ten questions

Plane angles at a vertex sum to:

360°

less than 360°

more

180°

Space diagonal of a $4 \times 6 \times 12$ cuboid:

12

14

22

$\sqrt{196}$

S_{total} of a right prism:

Ph

$2B + Ph$

Bh

$2Ph$

Volume of any prism:

$$Ph$$

$$Bh$$

$$\frac{1}{3}Bh$$

$$2Bh$$

Volume of a cylinder:

$$2\pi rh$$

$$\pi r^2 h$$

$$\pi rh$$

$$2\pi r(r+h)$$

Curved area of a cylinder:

$$\pi r^2$$

$$2\pi rh$$

$$2\pi r^2$$

$$2\pi r(r+h)$$

1 m³ in litres:

100

1000

10 000

1 000 000

A pipe's cross-section area:

$\pi(R-r)^2$

$\pi(R^2-r^2)$

πR^2

$2\pi R$

5 cm to the nearest cm has bounds:

4–6

4.5–5.5

4.9–5.1

4.95–5.05

Cavalieri's principle compares:

bases

slice areas at every height

perimeters

diagonals

04

Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Concept map	Tushunchalar xaritasi	Карта понятий
Surface area	Sirt yuzasi	Площадь поверхности
Volume	Hajm	Объём
Bounds	Chegaralar	Границы
Compound solid	Murakkab jism	Составное тело
Self-assessment	O‘z-o‘zini baholash	Самооценка
Target	Maqsad	Цель

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q4 asks for the curved area because:

it is easier

the question describes an open shape

it is always curved

no reason

A conversion mark is lost by:

using cm

mixing m and cm in one formula

using π

rounding

The volume bounds of a cuboid come from:

the stated values

all three extremes together

the largest edge

the average

Quarter III begins with:

- vectors
- the pyramid
- probability
- the cylinder

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Space diagonal of a $4 \times 6 \times 12$ cuboid

ANSWER 14

2. Surface area of a cube of edge 5

ANSWER 150

3. Volume of a prism, $B = 20$, $h = 9$

ANSWER 180

4. Volume of a cylinder, $r = 3$, $h = 8$

ANSWER 72π

5. 1 m^3 in litres

ANSWER 1000

6. Bounds of 7 cm to the nearest cm

ANSWER 6.5 and 7.5

7. Curved area, $r = 4$, $h = 10$

ANSWER 80π

Medium · the standard question

7 PROBLEMS

1. Angle of the $4 \times 6 \times 12$ space diagonal with the base

ANSWER $\approx 59.0^\circ$

2. Regular hexagonal prism, edge 5, height 11: total area

ANSWER $75\sqrt{3} + 330 \approx 460$

3. Trough 30/50 by 20 cm, 2.5 m long: litres

ANSWER 200

4. Cylinder holding 800 ml, height 10: radius

ANSWER ≈ 5.05 cm

5. Same: curved surface area

ANSWER ≈ 317 cm²

6. Pipe 2 m, $R = 9$, $r = 7$: metal volume

ANSWER $6400\pi \approx 20\,100$ cm³

7. Cuboid $5 \times 7 \times 9$ (nearest cm): volume bounds

ANSWER 248.6 to 391.9

Hard · stretch and reason

7 PROBLEMS

1. Same pipe: mass at 7.8 g/cm^3

ANSWER $\approx 157 \text{ kg}$

2. A silo, cylinder $r = 4$, $h = 12 \text{ m}$ with a hemisphere: volume

ANSWER $192\pi + \frac{128\pi}{3} \approx 737 \text{ m}^3$

3. A cylinder inscribed in a cube of edge 12: the ratio of volumes

ANSWER $\pi : 4$

4. A regular hexagonal prism of volume $540\sqrt{3}$ and edge 6: its height

ANSWER 10

5. A cuboid $3 \times 4 \times 5 \text{ m}$ to the nearest 0.1 m : volume bounds

ANSWER $\approx 58.4 \text{ to } \approx 61.7 \text{ m}^3$

6. Two similar prisms with volumes 8 and 125: the area ratio

ANSWER 4 : 25

7. A closed cylinder of volume 2000 cm^3 with least surface area: find r and h

ANSWER $r \approx 6.83$, $h \approx 13.66$

07 Homework

Homework — 4 tasks

Bring the concept map to the first lesson of Quarter III.

1. Rewrite in full every control-work question that lost a mark.
2. Finish the concept map with all six links written as sentences.
3. A cylindrical tank of radius 1.5 m and height 4 m : find its capacity in litres and its curved surface area.

4. Write your target for Quarter III in one checkable sentence, and date it.